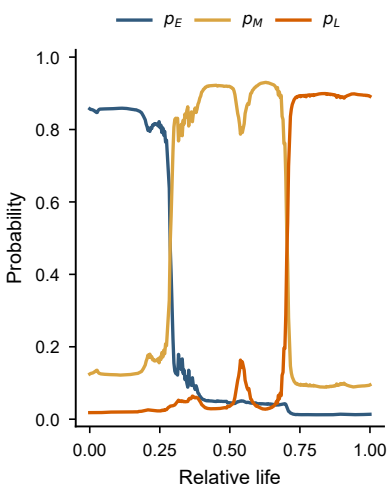
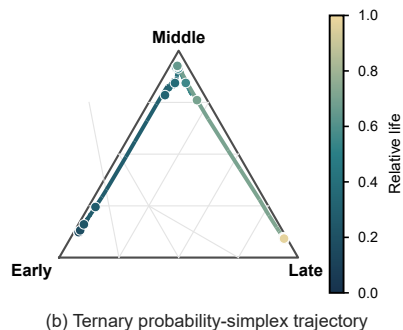
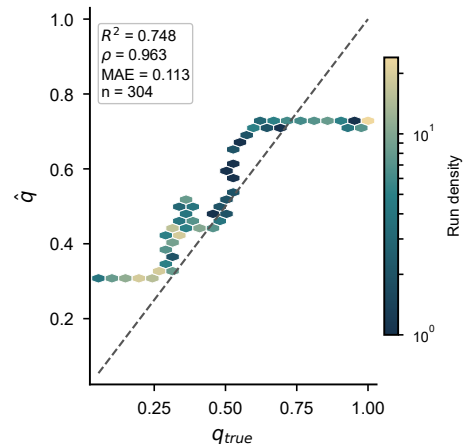




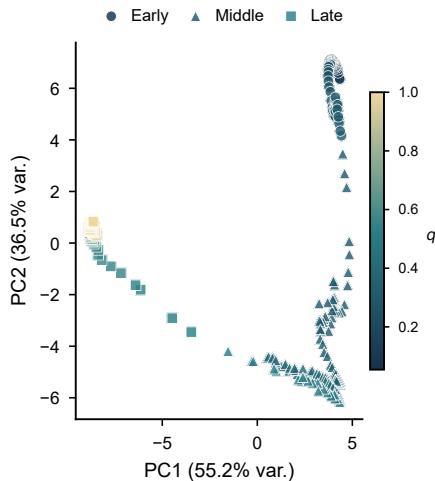
(a) Stage-aware probability evolution along tool life



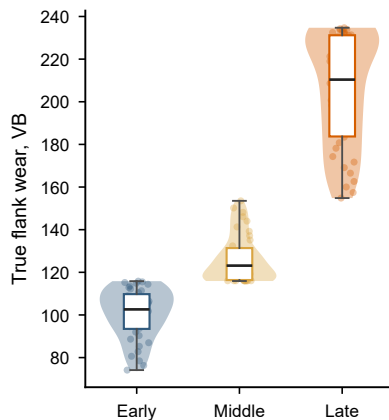
(b) Ternary probability-simplex trajectory



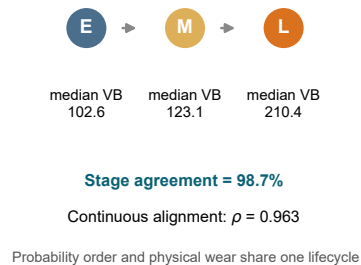
(c) Continuous q-estimation consistency



(d) Latent degradation manifold (deterministic PCA)



(e) Physical wear semantics by predicted stage



(f) Summary of degradation-semantic alignment